

Improving Generalization Through Hybrid Datasets in QA Tasks

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Abstract

Question answering (QA) models often fail to generalize beyond their training distribution due to dataset artifacts, which are lexical or structural patterns that models exploit as shortcuts rather than performing genuine reasoning. Prior work has shown that such artifacts can cause substantial performance drops when models are evaluated with adversarial examples. In this study, I investigate whether hybrid training data that combines standard QA examples with adversarial examples can improve generalization. Using ELECTRA-Small as the base model, I fine-tune on a mixture of 75% SQuAD and 25% SQuAD Adversarial examples, holding total training size constant. I evaluate this mixed-data model against a baseline trained solely on SQuAD across three QA benchmarks: SQuAD, SQuAD Adversarial, and TriviaQA. The mixed-data model improves performance on all datasets, yielding large gains on adversarial evaluation (EM +7.1, F1 +10.7) and moderate improvements on both in-domain SQuAD (+1.3 EM) and out-of-domain TriviaQA (+1.1 EM). Across all datasets, mixed-data training substantially reduces entity-type confusion errors, although it increases span-precision errors. These findings support prior evidence that adversarial exposure encourages deeper contextual reasoning.

1 Introduction

In the field of NLP, a significant problem has emerged: models have developed a tendency to learn data artifacts that interfere with their generalization capabilities. Data artifacts in NLP are systematic regularities in training data that models can exploit as shortcuts, allowing them to achieve seemingly high performance without learning the underlying linguistic or reasoning task. Models have been found to learn these patterns in question answering, natural language inference, reading comprehension, and other tasks.

In 2017, Jia and Liang found that many popular QA models were reliant on these dataset

artifacts, and when ‘distractors’ were inserted to manipulate these patterns, F1 scores fell from about 75% to 36%. The drop was largely attributed to semantic similarities among many questions in the SQuAD dataset, which was widely used for training and evaluating QA models.

In 2018, Kaushik and Lipton detailed further research into dataset artifacts in their paper "How Much Reading Does Reading Comprehension Require?". In their research, they asked what would happen if key pieces of information were removed from a reading comprehension task. They hypothesized that if key information was removed and the model still found the correct answer, it was relying on dataset artifacts rather than gaining a deeper understanding of the content. Kaushik and Lipton found that in 5 datasets, models given only the question, without context, achieved 20–40% of the accuracy of fully informed models. This implies that QA models can heavily rely on dataset artifacts present in their training data.

For my approach, I attempted to minimize the effects of data artifacts in a QA task by training a model on a mixture of baseline SQuAD data and SQuAD adversarial data. My hypothesis is that by including adversarial data during fine-tuning, the model will be able to generalize better to different question answering tasks.

2 Initial Approach

For a baseline model, I fine-tuned a pretrained model. To accomplish this, I used the pretrained ELECTRA-SMALL model. ELECTRA is a variant of BERT, a transformer-based language model. The ELECTRA-SMALL model has approximately 14 million parameters, with a reduced embedding size of 128 and four attention heads (Clark 2020).

For my baseline approach, I fine-tuned the model on the Stanford Question Answering Dataset (SQuAD). The dataset features 100,000 question-answer pairs derived from 536 Wikipedia articles across a broad range of topics. For this dataset, human annotators selected the correct answer by designating the span in the text where the answer appears (Rajpurkar et al. 2016).

My baseline model was fine-tuned on 8,000 examples from the SQuAD dataset for three epochs. Training used a limited number of examples due to hardware constraints (see limitations section). The model was evaluated on three datasets, using 1,000 examples from each.

This baseline model achieved an F1 of 55.9% on the SQuAD evaluation set. The baseline model, however, performed significantly worse on the other datasets. The model achieved an F1 of 9.6% on SQuAD Adversarial and 22.9% on TriviaQA questions. The performance drop suggests that, while the model performs well on SQuAD, it does not generalize well to other datasets.

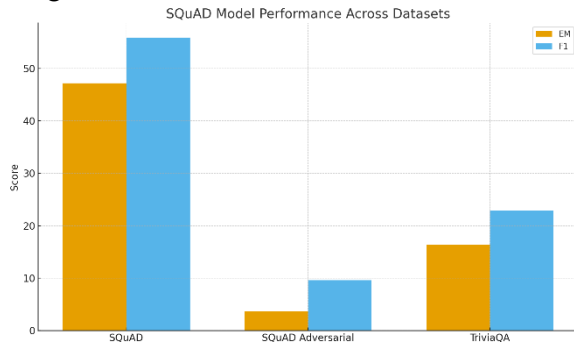


Figure 1: Baseline Model Performance Across Datasets.

3 Error Categories

To gather data on my approach, I categorized the errors into five distinct categories. I then utilized GPT-5.1 to sort the errors from each dataset into the five categories. The GPT model was given a prompt containing the error types, their descriptions, and the model's output.

To verify this approach, I randomly selected sections of the data and audited their labels. First, I randomly selected ten examples labeled as correct and verified that, for all ten, the model labeled them correctly. Then I audited 25 random questions marked as errors. I found that of these 25, only 1 was mislabeled. This example was labeled as wrong span – different sentence, when it would have been more accurate to label it as an entity type

confusion error. Overall, the GPT-5.1 model was accurate in categorizing these error types. Given the limited number of labeling mistakes it made (~4%), it is reasonable to consider these labels relatively accurate. The following error types were used for categorization.

- Entity Type Confusion
- Wrong Span – Same Sentence
- Wrong Span – Different Sentence
- Overly Long Span
- Overly Short Span

Each example was only given one label. In some cases, more than one label may be appropriate for an error. For example, an entity type confusion error could also be called a wrong span error. In these cases, priority was given to the entity type confusion label, as it provides more specific insight into the model's strengths and weaknesses. Similarly, while overly short or long spans could also be considered wrong span – same sentence errors, they were assigned the more specific short or long span label. The exception to this rule is the rare case where the span contains the correct answer but is excessively long at both ends; in these cases, it was categorized as a wrong span – same sentence error. Generally, priority was given to the more specific errors, with the hierarchy going entity type, overly short/long span, then wrong span.

An entity type confusion error is an error that occurs when the model predicts an answer with the wrong semantic type. This may involve a model predicting a location instead of a person, or a date instead of an organization. This often occurs when a model is searching for common grammatical patterns in the questions and answers. Generally, this error type can be summarized as the model not understanding what type of thing a question is asking for.

An example of this error from the TriviaQA dataset appears in the question "Who was the first person to sell one million records?". The gold label answer was "enrico caruso" but the model predicted "1902". This was the year Enrico Caruso sold a million records, but the model was confused about the type of entity it was looking for.

A wrong span error occurs when a model selects an incorrect span of words as the answer to a question. I differentiated this type of error into two types, same sentence and different sentence. If the

model selected an incorrect span in the correct sentence, it shows that it understands where in the text to find the answer but failed to extract the correct passage. If the model selects a span in the wrong sentence, it may indicate that it is far from the correct answer and has failed to find the relevant part of the text.

An example of a wrong span – different sentence error occurred in the question "What is the AFC short for?" in the SQuAD dataset. The model returned the answer "Super Bowl 50", when the answer was to be found in the following sentence "The American Football Conference (AFC) champion Denver Broncos defeated the National Football Conference (NFC) champion Carolina Panthers 24\201310 to earn their third or Super Bowl title." This error shows that the model did not identify even the approximate location of the relevant information.

An example of a wrong span – same sentence error also occurred in the SQuAD dataset. In this example, the question was "Super Bowl 50 decided the NFL champion for what season?" The gold label answer to this question was "2015" or "the 2015 season". The model made the prediction, "National Football League". The answers were in the same sentence: "Super Bowl 50 was an American football game to determine the champion of the National Football League (NFL) for the 2015 season." This error likely occurred because the model was able to locate the sentence in which the information was held, but the grammatical similarity of the position of "National Football League", as the subject likely confused the model into predicting it.

The final two types of errors are overly short spans and overly long spans. These errors occur when the model predicts a portion of the correct answer but either includes extra information or omits part of the correct answer.

An example of an overly long span is found in the model's responses to the TriviaQA dataset. The question was "The Black Hills lie between which two rivers?" The answer was "belle fourche and cheyenne" and the model predicted "Belle Fourche and Cheyenne rivers". While the model was not incorrect, it did not select the shortest span expected by the dataset. This is a limitation of the labeling scheme: a human can tell this is a correct answer, but an automatic categorization system cannot. Though this is a limitation, the category was still included due to these errors still showing

up in the automatic metric calculations used for quick model evaluation (see limitations section for more information).

The model's predictions from SQuAD Adversarial yielded an example of a short span error. The question was "How was teacher anxiety realized?" The model predicted "2012", while the gold label answer is "A 2012 study". While the model included part of the correct answer, it did not select a span that contained all the relevant information to constitute a full correct answer.

The five error categories provide a means to analyze the behavior of the model and its particular strengths. Using these, we can benchmark model modifications and observe how they affect performance on the question answering task.

4 Modified Approach

To improve generalization by minimizing data artifacts, I attempted an approach of mixing adversarial and regular SQuAD data into a hybrid dataset.

For simplicity, I will be referring to the original model as the "SQuAD model" and the modified model with adversarial data as the "mixed model".

Liu, Schwartz, and Smith (2019) researched similar methods. They proposed fine-tuning a model after training on ~100 adversarial examples, which they showed could drastically increase performance.

In my approach, I attempted to use a similar method. I constructed a hybrid dataset consisting of 75% examples from SQuAD and 25% examples from adversarial SQuAD.

Adversarial SQuAD is a dataset developed by Jia and Liang (2017), who created adversarial examples by inserting distracting sentences into context paragraphs. The distracting information would not have thrown off a human reader; instead, they focused on confusing models with word overlap and semantic proximity. In initial tests, this reduced F1 on particular models from ~75% to ~36%.

Using this hybrid dataset, I fine-tuned an ELECTRA-SMALL model with the same number of examples (8000) as my initial model. This ensures that any performance differences result from combining datasets rather than from using more training data.

5 Results

The two models were evaluated on their performance on three different QA datasets: SQuAD, SQuAD Adversarial, and a third dataset, TriviaQA (Joshi et al., 2017). The third dataset was included to test how well the models generalize to a dataset neither model had been exposed to during fine-tuning.

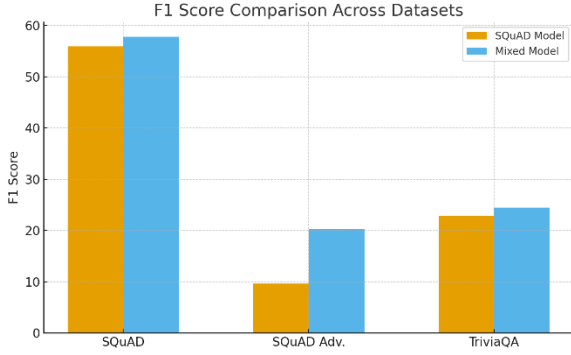


Figure 2: F1 Score Comparison Across Datasets

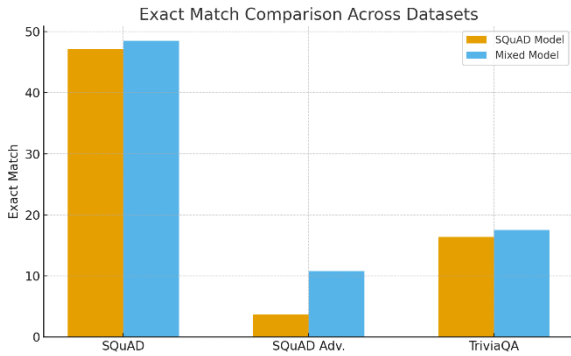


Figure 3: Exact Match Comparison Across Datasets

Overall, the mixed training model outperformed the SQuAD model in all cases. It was unsurprising that the mixed model performed vastly better on the SQuAD adversarial tasks, as the SQuAD adversarial examples had been included in its training data.

Interestingly, the mixed model slightly outperformed the base model on baseline SQuAD tasks. The mixed model achieved an exact match on 1.3% more examples and increased F1 by 1.9%. This was initially surprising, as the base model was trained only on the SQuAD dataset and had seen more examples from it.



Figure 4: Error Types on the SQuAD Dataset

Exploring the specific types of errors in the regular SQuAD dataset can help us begin to understand the results. The model trained on a mix of adversarial and regular data significantly reduced entity-type confusion errors. This error occurs when the model chooses the wrong type of object; for example, selecting a person instead of a location, or a date instead of a number. The decrease in errors of this type indicates that, when given adversarial examples, the model engages in deeper reasoning rather than looking for artifacts or semantic patterns in the text.

On the SQuAD adversarial task, the performance improved drastically from the base settings. This is to be expected, given data from SQuAD adversarial was part of the fine-tuning dataset for the mixed model. The model with a hybrid training set increased the exact match rate from 3.7% to 9.6% and boosted the F1 score from 9.6% to 20.3%.

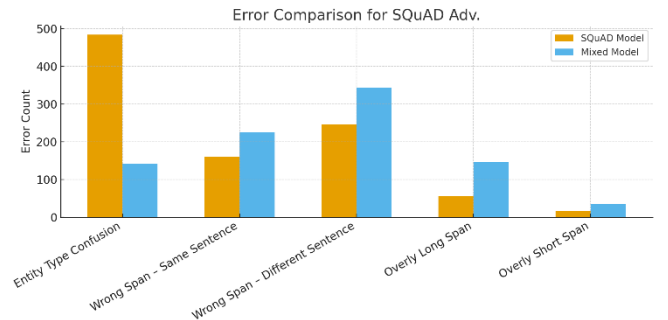


Figure 5: Error Comparison for SQuAD Adversarial

The specific types of errors showed patterns similar to those in the SQuAD dataset. The number of entity type confusion errors decreased from 348 to 142. The decrease in entity type confusion errors indicates that adding adversarial examples increased the model's ability to focus on the text's context rather than semantic patterns. As in SQuAD, the number of span errors of various types

increased, though the overall number of errors decreased.

The third task evaluated was TriviaQA. The goal with this dataset was to evaluate how well the modified approach improved generalization to a third task. The dataset features real trivia questions from trivia websites, and trivia leagues. It pairs those questions with Wikipedia pages and other web sources for context. It involves complex natural language phrasing. For this reason, it is a particularly good test of a model's ability to generalize to other tasks.



Figure 6: Error Types for TriviaQA

Overall, the mixed training data model slightly outperformed the model trained exclusively on SQuAD. The mixed training data model increased the exact match rate from 16.4% to 17.5%. It also increased F1 from 22.9% to 24.5%.

Despite the difference in overall performance being modest, the difference in the type of error was drastic. The mixed training data model reduced the number of entity type confusion errors from 334 to 141. It showed a performance drop in other error categories, with each increasing—particularly wrong sentence errors, which rose from 330 to 446.



Figure 7: Error Categories Across All Datasets

Overall, the modified training approach reduced the number of entity-type confusion errors across all datasets, while increasing errors

in other categories. This suggests that the mixed training data approach improved the ability of the model to interpret context, reducing its reliance on data artifacts.

Similar patterns of entity type errors declining alongside an increase in errors in precise calculations of spans occurred in the research of Liu, Schwartz & Smith (2019). In their research, they showed that using adversarial data improved their performance on semantic decision making but did have a cost in fine-tuned performance when selecting the correct span.

One example in which the mixed model performed better than the baseline is with the question, "Super Bowl 50 decided the NFL champion for what season?". The gold label answer for this question is either "2015" or "the 2015 season". The baseline SQuAD only model predicted "National Football League", which is an entity type confusion error. This is likely the result of a data artifact due to the wording of the passage. The context states "Super Bowl 50 was an American football game to determine the champion of the National Football League (NFL) for the 2015 season". The original model likely fixated on the grammatical pattern in the sentence, which mentions 'Super Bowl 50' determining the champion of something.

This entity type confusion error was not present in the mixed training data model, which correctly predicted "2015". The correct prediction seems to indicate that the model is better able to understand what it is predicting when it is given adversarial examples as well as baseline examples.

An example in which neither model could make the correct prediction is the question "What is the AFC short for?" The example is interesting because the context never explicitly states that AFC stands for "American Football Conference". Instead, the model would need to understand that the phrase "American Football Conference (AFC)" implies that "AFC" is an acronym.

Neither model successfully predicted the correct answer. The SQuAD model predicted "Super Bowl 50", while the mixed model predicted "Denver Broncos". Despite the incorrect answer, this may show deeper reasoning from the mixed model. The Denver Broncos are frequently mentioned as the team from the AFC, and a team is an organization which is the correct entity type. The model may have gained a greater understanding of the entity type of AFC from the mixed training dataset.

6 Limitations

One key limitation in this approach was the relatively small amount of data used for fine-tuning. The limited amount of training data was due to hardware constraints. The training was performed using an NVIDIA GTX 1650 Ti, with 4 GB of VRAM. With more powerful hardware, a more accurate baseline could be achieved. It is possible that a model fine-tuned on more data would have been stronger at baseline and therefore would not have shown as much improvement with a hybrid dataset.

Another limitation of this approach comes from the labeling scheme. In choosing a labeling scheme, I selected which error types to focus on. It is possible that if a different labeling scheme was examined, flaws in this approach might be revealed.

I also selected a fairly simple and objective labeling scheme. A more subjective labeling scheme would require deeper understanding of what went wrong. This would likely yield a greater understanding of the model’s performance. The reason I selected a simple scheme is that I could not label all 6,000 examples myself, so I needed to use an automated method. An LLM like the one I used (GPT-5.1) would likely be less accurate at applying a more complex labeling scheme than a simpler one. If more resources were available, such as human labelers, using a deeper labeling scheme, would likely have provided more insight.

One other limitation of this scheme is that some of the errors, particularly shorter or larger span errors, were in fact valid answers if it were to be evaluated by a human. For this reason, the numbers on the error count are slightly off. Due to the limited number of span errors and manual checking, I can verify that the number is minimal and does not change the overall result of the research.

7 Conclusion

Overall, I consider the result of this experiment to be a moderate success. While training with the adversarial data included in the fine-tuning dataset did create improvements, these improvements were most substantial in the adversarial data examples themselves and were

modest in the case of the TriviaQA set, which can be seen as a proxy for more generalized tasks.

In similar research Liu, Nelson, Schwartz, and Smith, utilized a related idea successfully in their paper *Inoculation by Fine-Tuning* (2019). Their approach differed as they fine-tuned exclusively on additional data. When they studied using their method on the SQuAD task, they found that fine-tuning on just 400 adversarial examples allowed their model to close 60% of the gap between performance on SQuAD and SQuAD adversarial tasks. However, they also observed a 7% drop in F1 on the original SQuAD datasets.

More experimentation with these methods may yield greater results. In the future, research into using multiple QA datasets or different ratios of adversarial and baseline data would help determine the practical efficacy of this approach for limiting the influence of dataset artifacts on NLP tasks.

References

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A Appendices

Exact Match Performance

	SQuAD	SQuAD Adversarial	TriviaQA
SQuAD Model	47.2	3.7	16.4
Mixed Model	48.5	10.8	17.5

Table 1: Exact match Performance Across Datasets & Models

F1 Performance

	SQuAD	SQuAD Adversarial	TriviaQA
SQuAD Model	55.9	9.6	22.9
Mixed Model	57.8	20.28	24.5

Table 2: F1 Performance Across Datasets & Models

SQuAD Model Error Types on SQuAD Questions	
Error Category	Count
Exact Match (Correct)	452
Entity Type Confusion	340
Wrong Span – Same Sentence	66
Wrong Span – Different Sentence	59
Overly Long Span	55
Overly Short Span	28
Total	1000

Table 3: Error Type Classifications for the SQuAD model on SQuAD questions

Mixed Model Error Types on SQuAD Questions	
Error Category	Count
Exact Match (Correct)	466
Entity Type Confusion	279
Wrong Span – Different Sentence	82
Wrong Span – Same Sentence	75
Overly Long Span	68
Overly Short Span	30
Total	1000

Table 4: Error Type Classifications for the mixed model on SQuAD questions

SQuAD Model Error Types on SQuAD Adversarial Questions	
Error Category	Count
Exact Match (Correct)	35
Entity Type Confusion	484
Wrong Span – Different Sentence	246
Wrong Span – Same Sentence	161
Overly Long Span	57
Overly Short Span	17
Total	1000

Table 5: Error Type Classifications for the SQuAD model on SQuAD Adversarial questions

Mixed Model Error Types on SQuAD Adversarial Questions	
Error Category	Count
Exact Match (Normalized EM)	108
Wrong Span – Different Sentence	343
Wrong Span – Same Sentence	226
Overly Long Span	146
Entity Type Confusion	142
Overly Short Span	35
Total	1000

Table 6: Error Type Classifications for the mixed model on SQuAD Adversarial questions

SQuAD Model Error Types on TriviaQA Questions	
Error Category	Count
Exact Match (Normalized EM)	164
Entity Type Confusion	334
Wrong Span – Different Sentence	330
Wrong Span – Same Sentence	87
Overly Long Span	84
Overly Short Span	1
Total	1000

Table 7: Error Type Classifications for the SQuAD model on TriviaQA questions

Mixed Model Error Types on TriviaQA Questions	
Error Category	Count
Exact Match (Normalized EM)	175
Wrong Span – Different Sentence	446
Entity Type Confusion	141
Wrong Span – Same Sentence	133
Overly Long Span	101
Overly Short Span	4
Total	1000

Table 8: Error Type Classifications for the mixed model on TriviaQA questions